

SYNERGY BETWEEN TRAJECTORIES AND HUMAN POSE FOR SOCCER

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ABSTRACT

Predicting the future body poses of professional athletes is a critical task for sports analytics, yet the complexity of broadcast footage presents a significant challenge for traditional vision-based forecasting. In this paper, we propose a novel spatio-temporal framework that jointly forecasts player trajectories in pitch coordinates and 2D body poses in normalized image coordinates. Our approach leverages the bidirectional synergy between movement and articulation: global trajectories serve as a physical prior to pose predictions, while body poses encode movement information that improves trajectory forecasting. Our model uses a velocity-based contrastive loss to regularize the latent space, ensuring motion-consistent predictions across different intensity regimes. Experiments on a large-scale proprietary soccer dataset and on the public WorldPose benchmark demonstrate that our model significantly outperforms pose-only baselines.

Index Terms— Sports Analytics, Trajectory Forecasting, Human Pose Estimation.

1. INTRODUCTION

The automated analysis of professional sports has undergone a significant transformation due to the availability of high-precision tracking and broadcast-based pose estimation [1, 2, 3]. In the high-stakes domain of professional soccer, predicting the future behavior of athletes is essential for tactical coaching and injury prevention [4, 5]. However, a significant gap remains in how the field treats the relationship between a player’s global movement and their local body articulation. Most existing research treats these as isolated streams: trajectory forecasting typically focuses on the social movement of entities across the pitch [6, 7, 8, 9, 10, 11, 12] while human pose forecasting focuses on the relative movement of joints in a localized vacuum [13, 14, 15, 16, 17]. Recent studies have begun to acknowledge the need for multimodality,

such as Social-Pose [18], Multi Trans-Motion [19] and MotionMap [20], which suggest that positional cues are vital for realistic motion prediction. By ignoring global trajectories, traditional models lose critical contextual priors that could resolve pose ambiguities in high-intensity scenarios.

In complex sports scenarios, like soccer with around twenty players in each dynamic scene, traditional 3D estimation models often struggle with depth ambiguities and occlusion; we argue that a highly precise, motion-aware 2D representation is the key to unlocking more robust 3D estimation. While recent work has explored cross-view completion [21, 22] to refine 3D understanding, these methods rely on having multiple views of the scene. We address these challenges by proposing a framework that leverages the synergy between pitch trajectories and articulated motion. Our architecture builds upon the successes of factorized attention mechanisms, which have proven effective in separating spatial and temporal dependencies in video tasks. Although recent literature has explored State-Space Models (SSMs) like Mamba [23] for their linear-scaling properties, we find that for the high-frequency transitions inherent in 3-second soccer clips, the perfect memory of Self-Attention Blocks (SAB) [24] remains the superior inductive bias. To further refine this representation, we introduce a *velocity-based contrastive* loss. By clustering players according to their physical momentum, we regularize the latent space to ensure that the forecasted poses are motion-consistent, effectively mitigating the issue of the model collapsing into average poses where predictions are static instead of high-intensity articulation.

To further regularize the learned representations, we draw inspiration from contrastive learning approaches that have demonstrated the value of clustering semantically similar samples in latent space. Barquero *et al.* [25] show that structuring the latent space according to behavioral modes leads to more diverse and realistic human motion predictions, while Perez *et al.* [26] demonstrate that prototype-based feature separation yields more discriminative representations in a detection context. Motivated by these findings, we introduce a velocity-based contrastive loss that clusters players according to their physical momentum, pulling together embeddings of

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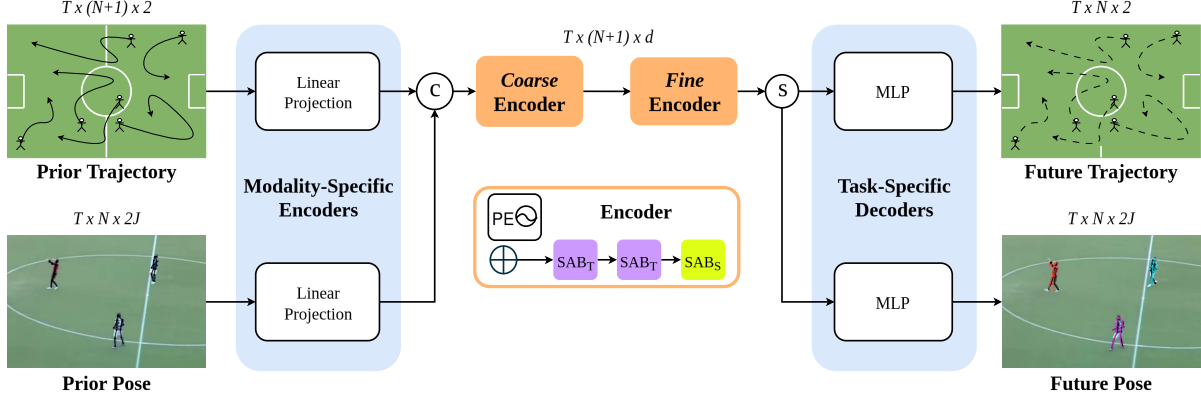


Fig. 1: Method Overview. Prior trajectories and poses are first projected independently into a shared latent space. The fused representation is processed by two stacked transformer encoders, denoted as coarse and fine. Each encoder consists of two temporal and a social SAB. The output of the final encoder is fed into task-specific heads implemented as independent MLPs.

players in similar motion regimes and pushing apart those with different intensities. This prevents the model from collapsing toward static average poses and encourages motion-consistent articulation across the forecast horizon. The contributions of this work are: 1) we present a transformer-based algorithm that jointly forecasts global trajectories and local 2D body poses with zero access to future context; 2) a contrastive objective that utilizes physical velocity to refine pose representations is introduced, ensuring they remain consistent with the player’s motion regime; 3) the synergy between trajectories and body poses with a comprehensive evaluation on a large-scale proprietary dataset of professional matches and on the public WorldPose dataset [27] is proved.

2. METHODOLOGY

Given a scene composed of $N + 1$ entities, including all players and the ball, and a temporal window consisting of T observed frames (2 seconds) followed by a prediction horizon of 1 second, the objective is to forecast the future motion trajectories of the players as well as their corresponding body poses with J joints. During the observation period, $t \in \{1, \dots, T\}$, the model receives two distinct input streams: player and ball trajectories, $\mathbf{x}_t \in \mathbb{R}^{(N+1) \times 2}$, expressed in global pitch coordinates, and normalized body poses, $\mathbf{p}_t \in \mathbb{R}^{N \times J \times 2}$, represented in 2D image coordinates. The pose representations are pre-normalized as joint-wise offsets relative to the center of each player’s bounding box. Operating on relative coordinates ensures invariance to global translation, camera motion, and variations in image resolution, thereby enabling the model to focus exclusively on articulated body motion.

The model is strictly causal, leveraging only the observed 2-second history to predict future trajectories and body poses for $t \in \{T + 1, \dots, T + T_f\}$, without access to future context or direct reference to the image frames. Although ball tra-

jectories are incorporated as an input modality to provide relevant game context, the forecasting objectives supervise exclusively player-specific motion and body articulation. The predicted poses are expressed in the same normalized coordinate system as the inputs, ensuring that the model captures the intrinsic dynamics of human motion independently of the player’s absolute location within the camera view.

2.1. Model

As depicted in Fig. 1, trajectories and poses are first projected independently into a shared latent space of dimension d using modality-specific fully connected layers. This yields two tensors of shape $T \times (N + 1) \times d$, which are concatenated along the feature dimension and then compressed back to $T \times (N + 1) \times d$ using Multilayer Perceptron (MLP). This operation enables early fusion while maintaining a fixed embedding size for subsequent processing.

The fused representation is processed by two stacked transformer encoders, denoted as coarse and fine stages [28]. Both encoders share identical architectural design. A temporal positional encoding is added to preserve frame ordering, while no positional encoding is applied across scene elements to enforce permutation equivariance with respect to player ordering. Each encoder consists of three self-attention blocks (SAB) [24]: two temporal attention blocks that mix information across frames independently for each entity, and one social attention block that models interactions among entities within the same frame. This factorized attention design allows the model to jointly capture individual motion dynamics and collective team behavior while remaining computationally efficient. We also explored replacing temporal self-attention blocks with Mamba-style state-space layers for sequence modeling [23], motivated by their linear-time complexity. While this variant provided faster training and in-

ference, it consistently underperformed self-attention in pose and trajectory forecasting accuracy. We therefore adopted temporal self-attention blocks.

The output of the final encoder is fed into two task-specific heads implemented as independent MLPs. The trajectory head outputs a tensor of shape $T \times N \times 2$, representing pitch-space positions. Only the last T_f frames are retained as the trajectory forecast. The pose head outputs a tensor of shape $T \times N \times 2J$, corresponding to the 2D joint coordinates for each player, again retaining only the future frames. Both heads, trained jointly with a composite loss function, operate on the same shared spatio-temporal representation, enabling consistent forecasting of motion and articulated pose.

2.2. Losses

We now introduce the losses we use to train our model. First, we consider a term to measure the error between predicted and ground-truth player trajectories in pitch coordinates as:

$$\mathcal{L}_{\text{traj}} = \frac{1}{NT} \sum_{n=1}^N \sum_{t=1}^T \|\hat{\mathbf{x}}_{n,t} - \mathbf{x}_{n,t}\|_2, \quad (1)$$

where $\hat{\mathbf{x}}_{n,t}$ and $\mathbf{x}_{n,t}$ are the predicted and ground truth positions for player n at frame t , respectively. In addition to supervising future motion, we enforce reconstruction of the observed trajectories too, effectively training the model in a sequence-to-sequence manner. This auxiliary supervision regularizes the temporal dynamics learned by the transformer and encourages smooth transitions across the observation–prediction boundary.

Similarly, a pose loss $\mathcal{L}_{\text{pose}}$ is applied to both observed and predicted frames as:

$$\mathcal{L}_{\text{pose}} = \frac{1}{NT} \sum_{n=1}^N \sum_{t=1}^T \left(\|\hat{\mathbf{p}}_{n,t,j}^x - \mathbf{p}_{n,t,j}^x\| + \|\hat{\mathbf{p}}_{n,t,j}^y - \mathbf{p}_{n,t,j}^y\| \right),$$

where $\hat{\mathbf{p}}_{n,t,j}$ is the prediction of the j -th joint for player n in frame t , and $\mathbf{p}_{n,t,j}$ is the ground truth. By supervising both the reconstruction of observed poses and the forecasting of the future ones, we enforce temporal consistency within the latent embeddings. The pose loss utilizes a coordinate-wise absolute aggregation before applying a l_2 -norm across all joints. This two-step approach ensures the loss remains sensitive to errors in both spatial dimensions independently, preventing the model from minimizing the objective by optimizing only a single coordinate axis. Applying an l_2 -norm over the joint set serves as a crucial regularizer, as it heavily penalizes large localized errors, such as those occurring in rapid limb motions, which might otherwise be masked by the accurate prediction of more stable joints like the torso or head. This design choice discourages degenerate solutions where the model converges toward a static average poses. Instead, it incentivizes the learning of fine-grained articulation

and high-intensity motion patterns essential for realistic soccer dynamics.

We also consider a contrastive loss applied to the latent embeddings produced by the transformer. It is motivated by the hypothesis that players exhibiting similar motion patterns, such as running, walking, or standing, share common pose dynamics. Using observed trajectories in pitch coordinates, we compute each player’s mean speed over the observation window and partition the N players into $K = 3$ velocity clusters $\{\mathcal{C}_1, \dots, \mathcal{C}_K\}$ via uniform speed quantization. Let $\mathbf{z}_n \in \mathbb{R}^d$ denote the ℓ_2 -normalized latent embedding of player n , and let $\mathcal{P}(n) = \{m \neq n \mid k(m) = k(n)\}$ be the set of players sharing the same velocity cluster as n . The contrastive loss is defined as:

$$\mathcal{L}_{\text{cont}} = -\frac{1}{N} \sum_{n=1}^N w_{k(n)} \log \frac{\sum_{m \in \mathcal{P}(n)} \exp(\mathbf{z}_n^\top \mathbf{z}_m / \tau)}{\sum_{m \neq n} \exp(\mathbf{z}_n^\top \mathbf{z}_m / \tau)}, \quad (2)$$

where $k(n)$ is the velocity cluster of player n , $\tau = 0.1$ is a temperature hyperparameter, and $w_{k(n)} = \frac{N}{|\mathcal{C}_{k(n)}|}$ is a cluster-frequency weight that corrects for imbalanced motion-regime distributions across the batch. The contrastive objective pulls together embeddings of players within the same motion cluster and pushes apart embeddings associated with different motion regimes. This regularization encourages motion-consistent representations and improves the model’s ability to capture subtle pose dynamics associated with different motion intensities.

Finally, our total loss is trained end-to-end by jointly supervising motion accuracy, articulated pose quality, and motion-consistent representation learning; being written as:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{traj}} \mathcal{L}_{\text{traj}} + \lambda_{\text{pose}} \mathcal{L}_{\text{pose}} + \lambda_{\text{cont}} \mathcal{L}_{\text{cont}}, \quad (3)$$

where $\{\lambda_{\text{traj}} = \lambda_{\text{pose}} = 1, \lambda_{\text{cont}} = 0.5\}$ are weight values.

3. EXPERIMENTAL EVALUATION

We present both quantitative and qualitative results to evaluate the effectiveness of our multimodal forecasting framework, focusing on the model’s ability to predict coherent global motion and precise local articulation over a 1-second future horizon. For the quantitative experiments, we provide four primary metrics. The trajectory metrics: Average Displacement Error (ADE), Final Displacement Error (FDE), and Maximum Error (MaxErr); all of them are computed in pitch coordinates and expressed in meters (m). ADE is defined in Eq. (1) as the mean Euclidean distance over all predicted frames. FDE measures the prediction accuracy at the final time step T_f such as $\text{FDE} = \frac{1}{N} \sum_{n=1}^N \|\hat{\mathbf{x}}_{n,T_f} - \mathbf{x}_{n,T_f}\|_2$. MaxErr represents the worst-case displacement throughout the predicted sequence, being defined by $\text{MaxErr} = \max_{t \in \{T, \dots, T_f\}} (\|\hat{\mathbf{x}}_t - \mathbf{x}_t\|_2)$. To evaluate the quality of the predicted pose, we use the Mean Joint Error (MJE) expressed in pixels as $\text{MJE} = \frac{\sum_{n=1}^N \sum_{t=T+1}^{T_f} \sum_{j=1}^J \|\hat{\mathbf{p}}_{n,t,j} - \mathbf{p}_{n,t,j}\|_2}{N \cdot (T_f - T + 1) \cdot J}$.

Complementary input	Roles	Pose Forecasting (px)		Trajectory Forecasting (m)	
		MJE↓		ADE↓ / FDE↓ / Max.Err.↓	
		Proprietary	WorldPose [27]	Proprietary	WorldPose [27]
✗	✓	1.2885	-	0.671 / 1.096 / 1.166	- / - / -
✗	✗	1.2119	2.5641	0.627 / 1.071 / 1.131	1.857 / 2.682 / 2.721
✓	✓	1.1818	-	0.677 / 1.114 / 1.177	- / - / -
✓	✗	1.1784	2.3226	0.573 / 1.024 / 1.074	1.566 / 2.555 / 2.622

Table 1: Bidirectional synergy of trajectory and pose modalities. Adding trajectory as a complementary input improves pose forecasting (MJE in pixels, left), while adding pose as a complementary input improves trajectory forecasting (ADE/FDE/MaxErr in meters, right). Team identity consistently degrades performance across both tasks. Results on WorldPose [27] validate cross-dataset generalization.

3.1. Dataset

In this work, we use two datasets, summarized below. **Proprietary Soccer Dataset.** Our main experiments use a proprietary soccer dataset containing 73 professional matches from a top-tier European league (2022–2023 season). We extract 13,237 clips of 3 seconds (75 frames at 25 fps) in which the ball remains in play, split into 10,896/1,082/1,259 samples for training/validation/testing. Splits are match-disjoint to promote generalization to unseen environments. The dataset includes annotated pitch-coordinate trajectories for all players ($N = 22$) and the ball. For body pose representation, we extract 2D skeletons with $J = 17$ using PoseTrack18 [1], normalizing joint positions relative to each player’s bounding box center to emphasize pose dynamics over image resolution and camera motion. **WorldPose Dataset.** We additionally evaluate on the public WorldPose dataset [27], which provides player trajectories and ground-truth body poses but no ball tracking. Following [27], we extract 3-second clips and use the standard train/validation/test splits.

3.2. Synergy of Trajectories and Poses

Table 1 provides a bidirectional contribution of trajectory and pose modalities across both datasets. These experiments form the key validation of our core hypothesis: that global and local motion representations are mutually beneficial priors.

Trajectories improve pose forecasting. The left half of Table 1 reports MJE under different input configurations. Providing global pitch trajectories as input consistently improves pose prediction accuracy on both datasets. This is consistent with our design motivation: a player’s global velocity constrains the set of plausible body configurations, providing a strong biomechanical prior that resolves pose ambiguities in high-intensity scenarios.

Poses improve trajectory forecasting. The right half of Table 1 shows the symmetric result. Access to the observed pose sequence consistently improves trajectory prediction across all metrics on both datasets. Body pose encodes the phase of a player’s movement cycle, whether player limbs

are moving faster would tell that player is accelerating, providing fine-grained information that complements the coarser positional history available to the trajectory head.

Team identity is detrimental. Across both tasks and both datasets, including player team membership as an input modality consistently degrades performance. Although one might expect team identity to provide useful context for modeling collective behavior, we argue the opposite: the social self-attention block already captures inter-player relationships geometrically, through proximity and relative velocity, without requiring explicit group labels. Introducing a discrete team label imposes a prior grouping that carries no geometric or biomechanical information, specifically for as short as 3-second windows, and risks introducing team-specific biases from the training matches that do not generalize.

To evaluate cross-dataset generalization, models pre-trained on our proprietary dataset are fine-tuned on WorldPose [27] training set. Since the latter does not provide team metadata, these experiments are conducted without the roles modality. The consistent performance benefit of each modality across datasets validates the robustness of our architecture.

3.3. Quantitative Evaluation

Table 2 compares two temporal modeling backbones with and without the proposed contrastive loss. Using SAB [24] as the temporal backbone consistently outperforms the Mamba-based variant across all metrics. While Mamba [23] offers efficient linear scaling for long sequences, our clips span only 75 frames, making this advantage negligible. More critically, Mamba’s recurrent state transitions introduce information decay that is particularly harmful for the high-frequency joint articulation patterns characteristic of athletic motion. SAB’s all-to-all attention allows every frame to directly attend to every other frame, which is essential for capturing the periodicity of gait cycles within short sequences.

The inclusion of the contrastive term in the total loss leads to a clear reduction in MJE . Figure 2 illustrates the effect qualitatively along the full spectrum of motion regimes:

TB	CL	ADE↓	FDE↓	Max.Err.↓	MJE↓
Mamba [23]	✗	1.03	1.46	1.62	1.52
Mamba [23]	✓	1.02	1.43	1.60	1.47
SAB [24]	✗	0.82	1.26	1.38	1.29
SAB [24]	✓	0.81	1.24	1.35	1.23

Table 2: Ablation of temporal backbone (TB) and contrastive loss (CL). Trajectory metrics (ADE, FDE, MaxErr) are in meters; pose accuracy (MJE) is in pixels.

Noise range (px)	MJE↓	
	Proprietary	WorldPose [27]
0	1.2294	2.9938
0–1	1.3640	3.1531
0–2	1.6367	3.5054
0–3	1.9332	3.5473

Table 3: Robustness to bounding box noise. MJE of our best model under increasing uniform pixel noise applied to bounding box centroids at inference time.

a player standing in the top left sequence, walking bottom left and running bottom right. By clustering entities based on their speed and pulling similar motion-regime embeddings together, the model learns a latent space that is more aware of the physical state. This regularization prevents the model from generating average or static poses.

3.4. Robustness to Bounding Box Noise

Real broadcast detections are frequently noisy or jittery across frames. Table 3 reports *MJE* under increasing levels of uniform pixel noise applied to bounding box centroids at inference time, where the noise range defines the maximum displacement in pixels. Because predicted poses are expressed as joint offsets relative to bounding box centers, upstream jitter propagates as a rigid shift of the entire skeleton rather than as distortion of the predicted articulation. Accordingly, performance degrades gracefully across both datasets, confirming that the normalized coordinate representation provides inherent robustness to imperfect upstream detections.

3.5. Qualitative Evaluation

For qualitative evaluation, we reconstruct absolute image-space poses by re-applying ground-truth bounding box centers to the predicted offsets. This reconstruction is used only for visualization and is not available during inference. Figure 2 shows four players from our proprietary dataset across 5 frames sampled 0.2 seconds apart. Ground-truth joints are shown in black and predictions in color. The first frame corresponds to the last observed frame before forecasting. The selected examples cover the motion regimes targeted

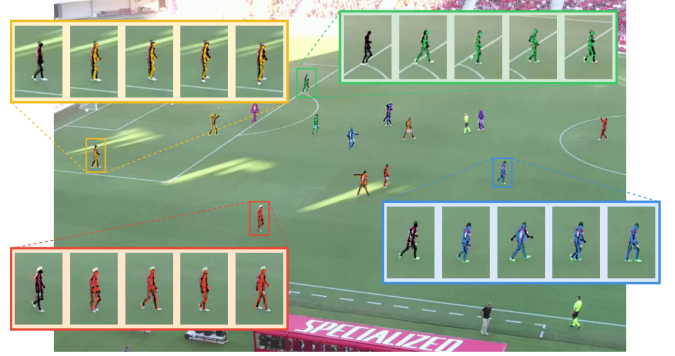


Fig. 2: Qualitative evaluation. Four players are zoomed in and shown across 5 frames, 0.2 seconds apart. One standing top left in yellow, one running bottom right in blue, other two in orange and green walking, covering the motion regimes induced by the contrastive loss. Ground-truth joints in black, predictions in color. The first frame corresponds to the last observed instant, subsequent frames are future predictions.

by the contrastive loss, and the model generates accurate, motion-consistent predictions in each case.

The figure also highlights the difficulty of pose forecasting in broadcast soccer footage, where players appear at low resolution and are affected by blur and occlusion. Despite these challenges, the model captures fine-grained articulation, such as leg-crossing dynamics during locomotion. Errors mainly occur for distal upper-limb joints in non-locomotive actions (e.g., pointing gestures), where global trajectory information provides limited physical constraints.

4. CONCLUSION

In this paper, we introduced a novel spatio-temporal framework for jointly forecasting player trajectories and body poses in professional soccer. Our results show a strong synergy between global and local motion: trajectories provide a biomechanical prior that improves pose forecasting, while body poses encode motion-phase information that enhances trajectory prediction. This complementary relationship is consistent across both our proprietary dataset and the public WorldPose benchmark, demonstrating the robustness of the approach. We further show that SABs outperform state-space models for modeling the high-frequency articulation dynamics of short athletic sequences, and that our velocity-based contrastive loss prevents convergence to static average poses, yielding motion-consistent predictions across different locomotion intensities. Finally, we believe the high-fidelity 2D forecasts produced by our model provide a strong basis for downstream 3D lifting tasks. Future work will focus on extending the framework to global 3D pose estimation in complex multi-entity settings using motion-consistent 2D priors and a shared reference system.

5. REFERENCES

- [1] M. Andriluka, U. Iqbal, E. Insafutdinov, L. Pishchulin, A. Milan, J. Gall, and B. Schiele, “Posetrack: A benchmark for human pose estimation and tracking,” in *CVPR*, 2018.
- [2] A. Agudo, “Piecewise Bézier space: Recovering 3D dynamic motion from video,” in *ICIP*, 2021.
- [3] Q. Xu, R. Cao, X. Shen, H. Du, S. Wang, and X. Yu, “M3GYM: A large-scale multimodal multi-view multi-person pose dataset for fitness activity understanding in real-world settings,” in *CVPR*, 2025.
- [4] Y. Shi, N. Ozeki, R. Yoshida, M. Inaji, K. Yagishita, and Y. Miyazaki, “Component replacement study of 3D human pose estimation models in real-world complex sports scenarios: Focusing on head impact events,” *ic-SPORTS*, 2024.
- [5] C. Yeung, K. Ide, and K. Fujii, “Autosoccerpose: Automated 3D posture analysis of soccer shot movements,” in *CVPR*, 2024.
- [6] M. Qi, J. Qin, Y. Wu, and Y. Yang, “Imitative non-autoregressive modeling for trajectory forecasting and imputation,” in *CVPR*, 2020.
- [7] Z. Fang, D. Hsu, G. H. Lee, and G. H. Lee, “Neuralized markov random field for interaction-aware stochastic human trajectory prediction,” in *ICLR*, 2025.
- [8] G. Capellera, A. Rubio, L. Ferraz, and A. Agudo, “Unified uncertainty-aware diffusion for multi-agent trajectory modeling,” in *CVPR*, 2025.
- [9] G. Capellera, L. Ferraz, A. Rubio, A. Alahi, and A. Agudo, “Jointdiff: Bridging continuous and discrete in multi-agent trajectory generation,” *ICLR*, 2026.
- [10] G. Capellera, A. Rubio, L. Ferraz, and A. Agudo, “Heteroscedastic diffusion for multi-agent trajectory modeling,” *TPAMI*, to appear, 2026.
- [11] W. Z. Teoh, “Coherent multi-agent trajectory forecasting in team sports with causaltraj,” *AAAIW*, 2026.
- [12] M. Peral, G. Capellera, L. Ferraz, A. Rubio, and A. Agudo, “Multi-modal soccer scene analysis with masked pre-training,” in *WACV*, 2026.
- [13] X. An, L. Zhao, C. Gong, N. Wang, D. Wang, and J. Yang, “Sharpose: Sparse high-resolution representation for human pose estimation,” in *AAAI*, 2024.
- [14] J. Zhang, Y. Yin, W. Yang, D. Ren, and D. Chen, “Posealign network for hybrid structure in 2D human pose estimation,” *SR*, vol. 15, no. 1, pp. 17117, 2025.
- [15] R. Han, M. Yi, W. Feng, F. Qi, and Y. Zhou, “Enhancing accuracy in dynamic pose estimation for sports competitions using hrpose: A hybrid approach integrating singlepose ai,” *AEJ*, vol. 127, pp. 200–213, 2025.
- [16] L. Zhong, F. Chen, B. Zheng, R. Feng, L. Zhang, and J. Wan, “DCT-diffpose: A lightweight diffusion model with multi-hypothesis for 3D human pose estimation,” *IEEE Access*, 2025.
- [17] C. Yeung, T. Suzuki, R. Tanaka, Z. Yin, and K. Fujii, “Athletepose3d: A benchmark dataset for 3d human pose estimation and kinematic validation in athletic movements,” in *CVPRW*, 2025.
- [18] Y. Gao, S. Saadatnejad, and A. Alahi, “Social-pose: Enhancing trajectory prediction with human body pose,” *TITS*, 2025.
- [19] Y. Gao, P.-C. Luan, and A. Alahi, “Multi-transmotion: Pre-trained model for human motion prediction,” in *CoRL*, 2024.
- [20] R. Hosseinienejad, M. Shukla, S. Saadatnejad, M. Salzmann, and A. Alahi, “Motionmap: Representing multi-modality in human pose forecasting,” in *CVPR*, 2025.
- [21] M. Armando, S. Galaaoui, F. Baradel, T. Lucas, V. Leroy, R. Brégier, P. Weinzaepfel, and G. Rogez, “Cross-view and cross-pose completion for 3D human understanding,” in *CVPR*, 2024.
- [22] R. Choudhury, K. M. Kitani, and L. A. Jeni, “Tempo: Efficient multi-view pose estimation, tracking, and forecasting,” in *ICCV*, 2023.
- [23] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” in *COLM*, 2024.
- [24] J. Lee, Y. Lee, J. Kim, A. Kosiorsek, S. Choi, and Y. W. Teh, “Set transformer: A framework for attention-based permutation-invariant neural networks,” in *ICML*, 2019.
- [25] G. Barquero, S. Escalera, and C. Palmero, “Belfusion: Latent diffusion for behavior-driven human motion prediction,” in *ICCV*, 2023.
- [26] M. Perez, A. Agudo, G. Dubbelman, and P. Jancura, “Class prototypical loss for enhanced feature separation in 3D object detection,” in *ITSC*, 2024.
- [27] T. Jiang, J. Billingham, S. Müksch, J. Zarate, N. Evans, M. R. Oswald, M. Polleyfeys, O. Hilliges, M. Kaufmann, and J. Song, “Worldpose: A world cup dataset for global 3D human pose estimation,” in *ECCV*, 2024.
- [28] G. Capellera, L. Ferraz, A. Rubio, A. Agudo, and F. Moreno-Noguer, “Transportmer: A holistic approach to trajectory understanding in multi-agent sports,” in *ACCV*, 2024.